



Questionnaire

“Did you hear any unusual sounds while playing the game?”

• Auditory Stimulus Recognition Check

- 1) Auditory Stimulus Recognition Check 1
“So, what specific sounds did you hear? If there were multiple, please describe them all.”
(We will not suggest specific types of sounds.)
“Is this the sound you heard?”
(Play the specific sound)
- 2) Auditory Stimulus Recognition Confirmation 2
“You mentioned hearing the sound of [XX] (the sound you identified in Auditory Stimulus Recognition Confirmation 1) during the VR play session just now, but we actually presented other auditory stimuli as well.”
“We played the sound of [XX] (the sound you did not identify in Auditory Stimulus Recognition Confirmation 1) during the VR play session just now. Do you remember hearing it?”

• Confusion with Reality

- 3) Confusion with Reality Confirmation
 - a) “Did you think (the sound you recognized) came from the VR headset? Or did you think it actually came from this laboratory?”
 - b) Yes, No “Please tell me why you answered that way.”
 - c) How would you rate the realism, fidelity, and “sense that it was coming from the real world” of the sound? Was it high? Or low?
 - i. (If low) Why did you feel it was low?

*(If you determined the sound came from the VR) What kind of sound did you think it was in the VR?

• Aggressiveness

- 4) Aggression Check
“When you heard the sound of [XX], did you feel as though you were being subjected to some kind of cyberattack from the VR app, the headset, or a third party?”
 - “When you heard the sound of ○○, did you sense any kind of malice?”
 - “What kind of malice did you sense?”
 - “Why did you sense that kind of malice?”
 - “When you heard the sound of ○○, did you feel any anxiety?”
 - “What kind of anxiety did you feel?”
 - “Why did you feel that sense of anxiety?”
 - “Besides anxiety, did you feel anything else—such as stress or

surprise? Anything at all?”

- **Visual Manipulation**

- 5) In one of the earlier questions, you mentioned that you recognized the sound (which you described as “unnatural”) as an unnatural sound. When you heard these sounds, did you turn your head toward the direction the sound came from? Please tell me which sounds made you turn your head and which ones didn’t. If you don’t remember, that’s fine.”
 - a) Yes
 - i) “Did you turn your head consciously? Or did you turn reflexively or unconsciously

- ?”
- ii) “Is there any reason why you turned your head when you heard that sound?”
 - b) No
 - i) “Why didn’t you turn around when you heard that sound?”

*Under what circumstances would you turn around? (While wearing VR)

* What kind of sound is effective?

● Further exploration

B. Sound Localization

- a. “It’s fine if you don’t remember exactly, but do you recall where the unnatural sounds you heard during this experiment seemed to be coming from?”
- c. Regarding the sounds for which clear head tracking was observed during the experiment,
 - a. “We played the sound of [sound] about [X] minutes into the game, and at that moment, from my perspective, it looked like you turned your head—do you remember that?”
- D. “By the way, do the notifications on the mobile devices you use regularly, such as your cell phone, come with a vibration? (If both, please give a rough estimate of the ratio.)”
 - a. (Check for any correlation between “Yes” and “No” answers)
- E. “Do you think that if the unnatural sounds used in this experiment were to occur in your daily life, you would find yourself looking in that direction?”
 - a. Specifically, which sounds do you think would elicit a reaction? If there are multiple, please list them all.
 - b. (If the subject answered “Yes” but did not react during the experiment) “Is there any reason why you didn’t react during this experiment?”

- 6) “(Regarding the sound newly identified in Auditory Stimulus Recognition 2) In response to the very first question—‘Did you hear any unusual sounds while playing the game?’—you did not mention hearing the sound of ○○. What do you think were the reasons you did not answer or could not recall it?”
- 7) “(Regarding sounds you did not recognize in Auditory Stimulus Recognition 2) This is a difficult question to answer, so you don’t need to force an answer. You mentioned that you were completely unable to recognize the sound of ○○. What do you think were the factors that caused you to fail to recognize that sound?”
- 8) Checking the Impact on the Task: “We presented you with five types of unnatural auditory stimuli: ‘vibration sounds,’ ‘the sound of breaking glass,’ ‘human voices,’ ‘the sound of clearing game blocks,’ and ‘a baby crying.’ Did these have any impact on your gameplay experience, concentration, or other aspects of your gameplay?”

Follow-up Questions

- “What specific effects did you notice?”
 - “Why do you think those effects occurred?”

- “Were there any sounds that had a particularly strong impact?”
 - “Why do you think that sound had such a strong effect?”
- “Conversely, was there a sound that had the least impact?”
 - “Why was the impact of that sound so small?”

Question Flow

In response to the question, “Did you hear any unnatural sounds while playing the game?”

- **Yes** → Go to Auditory Stimulus Recognition 1
 - If all items are checked in Auditory Stimulus Recognition 1, proceed to Questions 3, 4, and 5
 - Checked some items in Auditory Stimulus Recognition 1
For sounds that were not checked, proceed to Auditory Stimulus Recognition 2
 - Checked all items in Auditory Stimulus Recognition 2
 - Ask Questions 3, 4, and 5 for the sounds recognized in Auditory Stimulus Recognition 1 and 2
 - Ask Question 6 for sounds newly recognized in Auditory Stimulus Recognition 2
 - If there are sounds that were not checked after Auditory Stimulus Recognition 1 and 2
 - Ask Questions 3, 4, and 5 for the sounds recognized in Auditory Stimulus 1 and 2
 - Ask Question 6 about the sounds newly recognized in Auditory Stimulus 2
 - Question 7 regarding sounds that were not recognized
 - Questions 3, 4, and 5 regarding sounds recognized in Auditory Stimulus 1 but not newly checked in Auditory Stimulus Recognition 2
Ask Question 7 regarding sounds that were not recognized
 - If all were checked in Auditory Stimulus Recognition 1, proceed to Auditory Stimulus Recognition 2
 - Questions 3, 4, 5, and 6 for sounds newly checked in Auditory Stimulus Recognition 2
 - Question 7 for sounds that were not recognized
- **No.** Proceed to Auditory Stimulus Recognition 2
 - If all were checked in Auditory Stimulus Recognition 2, ask Questions 3, 4, 5, and 6
 - Checked some items in Auditory Stimulus Recognition 2
Questions 3, 4, 5, and 6 for the sounds that were checked
 - Question 7 regarding sounds that were not

- checked
Go to Question List ⑦ for sounds
that were not checked in
Auditory Stimulus Recognition 2